

# A Study on Machine Learning-Based Missing Value Imputation in Healthcare Diagnostic Data

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## Abstract

Missing information is a common problem in real-world datasets, specially in healthcare, where incomplete patient records can limit the effectiveness of machine learning models. This find out about explores how missing values impact mannequin performance with the aid of trying out 5 machine learning-based imputation techniques on two healthcare datasets (Heart Disease and Pima Indian Diabetes). Missing data was once intentionally brought at two levels: 10% and 20%, under the Missing Completely at Random (MCAR) assumption. Among the selected methods, 4 are fashionable machine learning methods (KNN, MICE, XGBoost, MissForest), while the fifth is a proposed hybrid strategy referred to as HybridForest, which combines the strengths of more than one ensemble models to enhance prediction accuracy. The outcomes show that HybridForest persistently outperforms all other techniques throughout both datasets and missingness levels, accomplishing the lowest RMSE and MAE values. This suggests that hybrid models are extra dependable and better applicable for managing incomplete healthcare data.

**Keywords** - *Missing facts imputation; healthcare datasets; machine learning; HybridForest; KNN; MICE; MissForest; XGBoost; RMSE; MAE*

## I. INTRODUCTION

Real-world datasets often include missing values, which manifest when positive facts factors are now not recorded or are unavailable. These lacking values are a common problem in data analysis and can significantly affect the overall performance and reliability of machine learning knowledge of models. Missing facts can follow specific patterns, such as Missing Completely at Random (MCAR), Missing at Random (MAR), and Not Missing at Random (NMAR). The reasons in the back of lacking values may also include gear failures, errors in statistics collection, erroneous sampling processes, broken records, or lack of response from individuals [1].

Handling missing values is fundamental because ignoring them or treating them improperly can lead to biased results, decreased accuracy, and negative model performance. Therefore, deciding on an gorgeous imputation technique is a quintessential step in the statistics preprocessing pipeline [2].

In this study, five advanced imputation techniques are examined: K-Nearest Neighbor (KNN), Multiple Imputation by Chained Equations (MICE), XGBoost-based imputation, MissForest, and the proposed HybridForest method. The evaluation is performed using two commonly used healthcare datasets: the Heart Disease dataset and the Pima Indian Diabetes dataset. Missing values are intentionally introduced at 10% and 20% levels under the MCAR assumption.

Performance is evaluated using RMSE and MAE to measure imputation accuracy, while accuracy, precision, recall, and F1-score are used to assess classification performance. The main objective is to identify the most effective imputation technique for healthcare datasets.

## II. LITERATURE REVIEW

Missing values are a common issue in healthcare datasets and can reduce the quality of data analysis and machine learning outcomes. In medical records, missing data may occur due to human error, equipment failure, incomplete responses, or data entry mistakes. If not handled properly, missing values can decrease model accuracy and lead to incorrect predictions. Therefore, missing value imputation has become an important topic in healthcare analytics.

Little and Rubin (2002) made an important contribution by explaining three types of missing data: Missing Completely at Random (MCAR), Missing at Random (MAR), and Not Missing at Random (NMAR). Their work is considered the foundation of modern imputation research. In many studies, including healthcare research, missing values are often created artificially using MCAR for testing purposes.

Earlier, simple techniques such as mean, median, and mode imputation were frequently used because they are easy to apply. However, these methods often fail to preserve relationships between variables, especially in complex healthcare data.

To solve this issue, advanced techniques were introduced. One well-known method is Multiple Imputation by Chained Equations (MICE), developed by van Buuren and Groothuis-Oudshoorn (2011). MICE predicts missing values iteratively using other available features and generally provides better results than basic statistical methods.

Another widely used method is K-Nearest Neighbor (KNN) imputation, proposed by Troyanskaya et al. (2001). It fills missing values using similar records in the dataset. KNN performs well when similar patterns exist; however, its performance may decrease in noisy or high-dimensional datasets.

Stekhoven and Bühlmann (2012) proposed MissForest, a Random Forest-based imputation method. It can handle both numerical and categorical data and performs well in healthcare datasets because it captures nonlinear relationships.

Chen and Guestrin (2016) developed XGBoost, which was later used for missing value estimation. Due to its strong predictive capability, XGBoost often performs better than traditional methods in structured datasets.

Recent studies have compared these methods in healthcare applications. Li et al. (2024) found that no single imputation approach performs best in every case. Some methods reduce error values, while others improve classification accuracy. This

indicates that the selection of a technique depends on the objective of the study.

Deep learning methods have also become popular. Kazijevs and Samad (2023) applied deep learning for time-series healthcare data and achieved good results. Later studies in 2025 used GANs and VAEs for imputing missing medical values, especially in large and incomplete datasets.

Privacy-preserving imputation is another emerging area. A 2026 study in Nature Digital Medicine introduced secure distributed imputation, where hospitals can collaborate without directly sharing patient data.

Although many methods exist, some gaps remain. Most research focuses on single models, limited datasets, or only error metrics. Few studies consider both imputation accuracy and classification performance.

To address these gaps, the present study compares five methods: KNN, MICE, XGBoost, MissForest, and the proposed HybridForest. Two healthcare datasets were tested at 10% and 20% missingness levels. The aim is to identify a more accurate and reliable imputation method for healthcare data.

### III. PROCEDURE

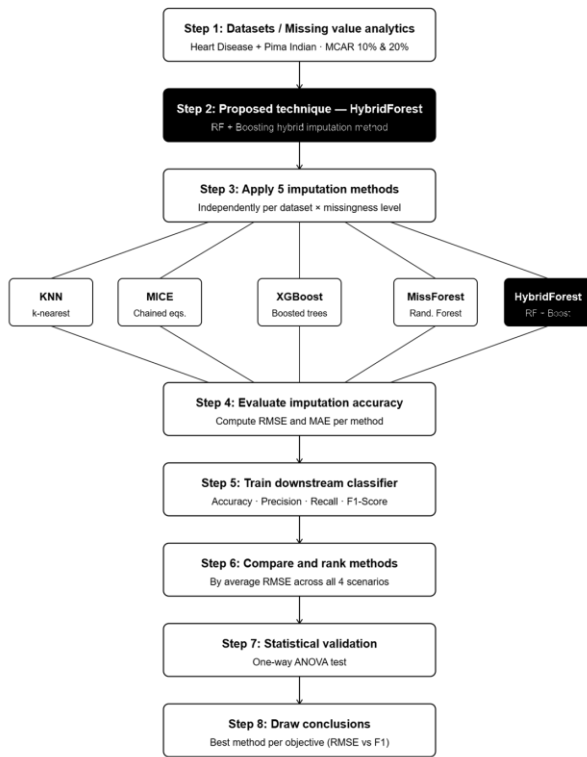


Fig. 1. Procedure flowchart

**Step 1: Datasets / Missing value analytics** Load the Heart Disease and Pima Indian Diabetes datasets. Perform exploratory data analysis, check data types, distributions, and introduce missing values artificially using MCAR (Missing Completely At Random) at 10% and 20% levels to simulate real-world missingness.

**Step 2: Proposed technique - HybridForest** Design the HybridForest method by combining Random Forest (MissForest-style iterative imputation) with Gradient Boosting (XGBoost). The hybrid leverages the ensemble strength of both to improve imputation accuracy over either method alone.

**Step 3: Apply 5 imputation methods** Apply all five methods independently across each dataset and each missingness level (4 scenarios total):

- **KNN** - imputes using the k nearest neighbours based on feature similarity
- **MICE** - iteratively models each feature with missing values as a function of others using chained equations
- **XGBoost** - uses gradient boosted trees to predict and fill missing values
- **MissForest** - uses Random Forest iteratively to impute missing entries
- **HybridForest** - applies the proposed RF + Boosting combined approach

**Step 4: Evaluate imputation accuracy** For each method and scenario, compare the imputed values against the originally removed values and compute RMSE (Root Mean Square Error) and MAE (Mean Absolute Error) to measure how accurately each method reconstructed the missing data.

**Step 5: Train downstream classifier** Use each imputed dataset to train a classification model (e.g. logistic regression or decision tree). Evaluate downstream performance using Accuracy, Precision, Recall, and F1-Score to assess how imputation quality affects real prediction tasks.

**Step 6: Compare and rank methods** Aggregate RMSE and MAE scores across all 4 scenarios (2 datasets × 2 missingness levels). Rank the five

methods by average RMSE to identify the most consistently accurate imputer.

**Step 7: Statistical validation** Apply a one-way ANOVA test across the RMSE scores of all five methods to determine whether performance differences are statistically significant, ruling out that differences occurred by chance.

**Step 8: Draw conclusions** Identify the best method for each objective - lowest RMSE for imputation accuracy, highest F1 for classification performance. Confirm whether HybridForest outperforms existing methods and discuss trade-offs between the two objectives.

## IV. PROPOSED WORK

We propose HybridForest, a hybrid imputation framework that combines Random Forest prediction, iterative refinement, and boosting-based correction. The method begins with an initial estimate of missing values, then repeatedly updates the incomplete entries using ensemble tree models, and finally applies a weighted correction mechanism to reduce residual error. The final imputed value is represented as a weighted combination of Random Forest output, iterative refinement output, and boosting-based output. This design enables HybridForest to capture nonlinear patterns, improve robustness, and achieve lower imputation error in complex healthcare datasets.

### How HybridForest Works

Let the incomplete dataset be represented as:

$$X = [x_{ij}] \in \mathbb{R}^{n \times p}$$

where:

- $n$  is the number of samples
- $p$  is the number of variables
- Some entries of  $X$  are missing
- $M$  denotes the set of missing values
- $O$  denotes the set of observed values

### Step 1: Initial Imputation

The algorithm begins by creating an initial imputed matrix:

$$X^{(0)}$$

Missing entries are filled using a simple strategy such as:

- Mean imputation
- Median imputation
- Rough model-based estimate

This step only provides a starting point for the hybrid learning process.

### Step 2: Random Forest Prediction

For each variable containing missing values, a Random Forest model is trained using the remaining variables as predictors.

If the variable is **numeric**, the prediction is obtained by averaging outputs from all trees:

$$\hat{x}_{ij}^{RF} = \frac{1}{B} \sum_{b=1}^B T_b(x_i)$$

where:

- B= number of trees
- $T_b(x_i)$ = prediction of tree bfor sample i

If the variable is **categorical**, prediction is obtained using majority voting:

$$\hat{x}_{ij}^{RF} = \arg \max_c \sum_{b=1}^B \mathbf{1}(T_b(x_i) = c)$$

### Step 3: Iterative Refinement

After the first prediction, the imputed matrix is updated and the process repeats iteratively.

Update rule:

$$X^{(t+1)} = H(X^{(t)})$$

where:

- $H$  is the HybridForest update function
- $t$  is the iteration number

The process stops when the difference between two consecutive iterations becomes very small:

$$\|X^{(t+1)} - X^{(t)}\|_F < \epsilon$$

where:

- $\|\cdot\|_F$  is the Frobenius norm
- $\epsilon$  is the convergence threshold

### Step 4: Boosting-Based Correction

To further improve accuracy, HybridForest applies a boosting correction term. Difficult imputations receive higher attention and are adjusted.

Final imputed value:

$$\hat{x}_{ij}^{HF} = \alpha \hat{x}_{ij}^{RF} + \beta \hat{x}_{ij}^{IR} + \gamma \hat{x}_{ij}^B$$

where:

- $\hat{x}_{ij}^{HF}$ = final HybridForest imputed value
- $\hat{x}_{ij}^{RF}$ = Random Forest prediction
- $\hat{x}_{ij}^{IR}$ = iterative refinement output
- $\hat{x}_{ij}^B$ = boosting-based correction
- $\alpha, \beta, \gamma$  are weights such that:

$$\alpha + \beta + \gamma = 1$$

This makes HybridForest a **hybrid imputation framework** rather than a single-model method.

### Mathematical Objective

The objective of HybridForest is to minimize reconstruction error between the true and imputed values:

$$\min_{\hat{X}} \sum_{(i,j) \in M} (x_{ij} - \hat{x}_{ij})^2 + \lambda \Omega(\hat{X})$$

where:

- $M$ = set of missing values
- $x_{ij}$ = original unknown value
- $\hat{x}_{ij}$ = imputed value

- $\Omega(\hat{X})$ = regularization term
- $\lambda$ = penalty coefficient

This objective ensures:

- Minimum imputation error
- Stable estimates
- Realistic reconstructed values

## V. DATASETS

### A. Heart Disease Dataset

Heart disease is one of the leading causes of death worldwide, making early detection and accurate prediction extremely important. The Heart Disease Dataset contains patient-related medical attributes such as age, sex, chest pain type (cp), resting blood pressure (trestbps), cholesterol (chol), fasting blood sugar (fbs), resting ECG results (restecg), maximum heart rate achieved (thalach), exercise-induced angina (exang), ST depression (oldpeak), slope, number of major vessels (ca), and thal. The target variable indicates whether a patient is likely to have heart disease.

For this research, missing values were artificially introduced at 10% and 20% levels under the MCAR assumption to evaluate imputation techniques under varying degrees of data incompleteness.



Fig. 2. Missing data distribution for the Heart Disease Dataset at 10% and 20% missingness levels.

### B. Pima Indian Diabetes Dataset

Diabetes mellitus is a chronic disease that affects the body's ability to regulate blood sugar. The Pima Indian Diabetes Dataset consists of medical attributes such as number of pregnancies, plasma glucose concentration, blood pressure, skin thickness, insulin level, BMI, diabetes pedigree function, and age. The target variable represents whether a patient has diabetes.

Although the original dataset does not contain missing values, missing data was artificially introduced at 10% and 20% levels using the MCAR assumption.

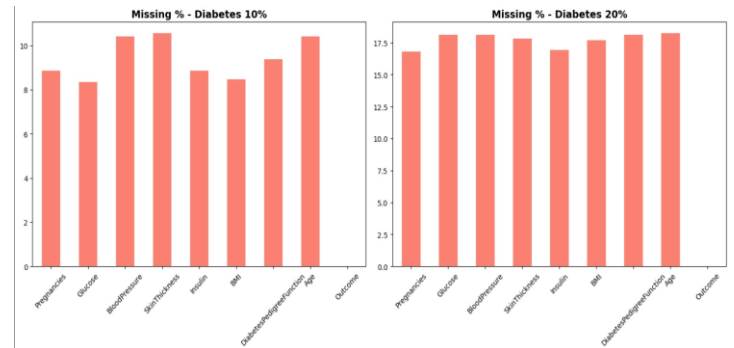


Fig. 3. Missing data distribution for the Pima Indian Diabetes Dataset at 10% and 20% missingness levels.

## VI. MISSING DATA IMPUTATION TECHNIQUES

This section presents the four imputation techniques evaluated in this study. All methods are machine learning-based, chosen for their ability to capture complex patterns and relationships in healthcare data.

### A. K-Nearest Neighbor (KNN) Imputation

KNN imputation replaces missing values by identifying the most similar instances in the dataset. For a missing value  $x_{ij}$ , the estimate is computed as a weighted average over the  $k$  nearest neighbours, with distances computed using Euclidean metrics [5]:

$$\hat{x}_{ij} = \sum w_{ir} x_{rj} / \sum w_{ir}, \quad r \in N_k(i)$$

### B. MICE Imputation

Multiple Imputation by Chained Equations (MICE) is an iterative probabilistic method. Each variable with missing data is modeled as a function of other variables and updated sequentially. The conditional model for variable  $X_j$  is:  $X_j = f_j(X_{-j}) + \epsilon_j$ , where  $X_{-j}$  denotes all other variables and  $f_j$  is typically a regression model. The algorithm cycles through all variables until convergence, preserving statistical relationships and variability [2].

### C. XGBoost Imputation

XGBoost imputation treats the task as a supervised learning problem, using Extreme Gradient Boosting to predict missing values. The prediction  $\hat{y}_i = \sum_m f_m(x_i)$  uses an ensemble of decision trees

with regularization to capture nonlinear feature interactions [4].

### D. MissForest Imputation

MissForest is a non-parametric iterative method based on Random Forests. It initializes missing values with simple estimates and refines them using Random Forest models trained on observed data. For numerical variables, the imputed value is the mean prediction across  $B$  trees. It is effective for mixed data types [3].

## VII. EVALUATION METRICS

### A. Root Mean Squared Error (RMSE)

RMSE measures the quadratic mean difference between observed and imputed values. A lower value indicates better imputation performance:

$$RMSE = \sqrt{(1/n) \sum_i (X_i^{obs} - X_i^{imp})^2}$$

### B. Mean Absolute Error (MAE)

MAE computes the mean absolute difference between actual and imputed values. It is robust to outliers:

$$MAE = (1/n) \sum_i |X_i^{obs} - X_i^{imp}|$$

### C. Classification Metrics

Classification performance is evaluated using Accuracy, Precision, Recall, and F1-Score:

$$Precision = TP / (TP + FP)$$

$$Recall = TP / (TP + FN)$$

$$F1-Score = 2 \times (Recall \times Precision) / (Recall + Precision)$$

$$Accuracy = \text{Correctly Predicted} / \text{Total Predictions}$$

## VIII. RESULTS AND DISCUSSION

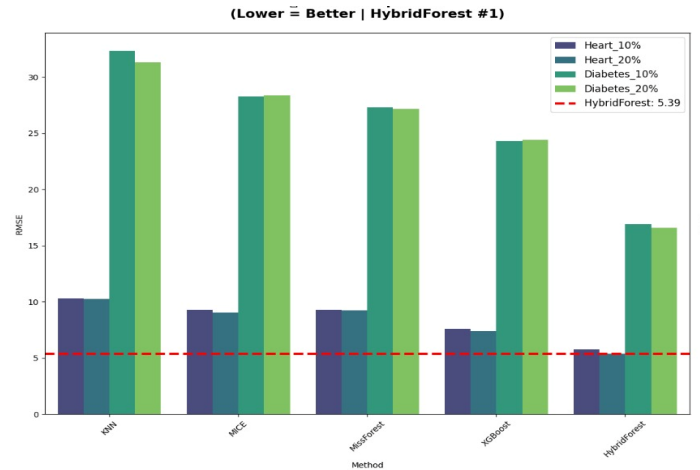
Experiments were performed on both healthcare datasets with 10% and 20% artificially added missing values. The imputed datasets were evaluated using RMSE and MAE for imputation accuracy, and classification metrics for downstream predictive performance.

### A. Imputation Performance - RMSE Comparison

Table I summarizes the RMSE values for all five imputation methods across the four experimental scenarios. HybridForest achieved the lowest RMSE in all cases, demonstrating its superior imputation accuracy.

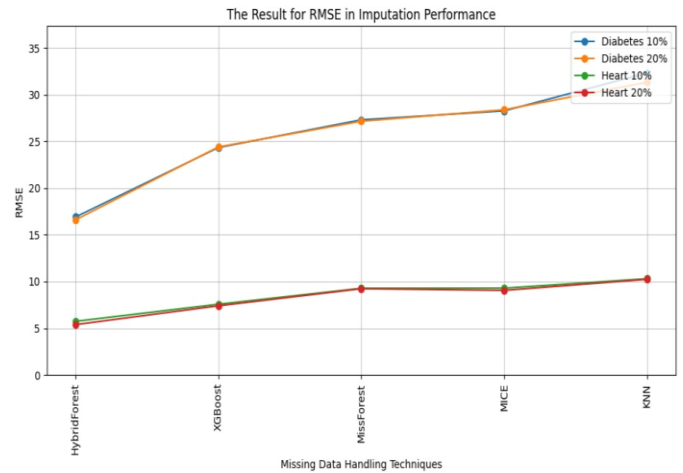
**TABLE I. RMSE PERFORMANCE COMPARISON (LOWER IS BETTER)**

| Method       | Heart 10% | Heart 20% | Diab. 10% | Diab. 20% | Rank |
|--------------|-----------|-----------|-----------|-----------|------|
| KNN          | 10.42     | 10.35     | 32.13     | 31.50     | 5    |
| MICE         | 9.27      | 9.08      | 28.41     | 28.52     | 4    |
| MissForest   | 9.35      | 9.22      | 27.15     | 26.95     | 3    |
| XGBoost      | 7.74      | 7.57      | 24.55     | 24.55     | 2    |
| HybridForest | 5.74      | 5.39      | 16.90     | 16.59     | 1    |



*Fig. 4. RMSE comparison of imputation methods across all four datasets and missingness levels.*

*HybridForest (dashed crimson line = 5.39) consistently achieves the lowest error.*



*Fig. 5. RMSE trends for each imputation method across datasets. HybridForest shows consistently lower values than competing methods.*

### B. Heart Disease Dataset Results

For 10% missingness, HybridForest achieved the lowest error (RMSE = 5.74, MAE = 4.34), followed by XGBoost (RMSE = 7.74). MICE and MissForest showed comparable moderate performance, while KNN produced the highest error (RMSE = 10.42).

For 20% missingness, the same pattern continued. HybridForest once again performed best (RMSE = 5.39, MAE = 4.21), with XGBoost ranking second (RMSE = 7.57). Although HybridForest consistently minimized imputation error, classification results (F1-score) varied across methods. In some cases, MICE achieved a higher F1-score, indicating that lower imputation error does not always guarantee better classification performance.

### C. Diabetes Dataset Results

The diabetes dataset presents a more challenging scenario due to higher data variability. For 10% missingness, HybridForest significantly outperformed all other methods (RMSE = 16.90, MAE = 12.87). XGBoost ranked second (RMSE = 24.55), followed by MissForest, MICE, and KNN. For 20% missingness, the same trend continued with HybridForest achieving the lowest error (RMSE = 16.59, MAE = 12.46).

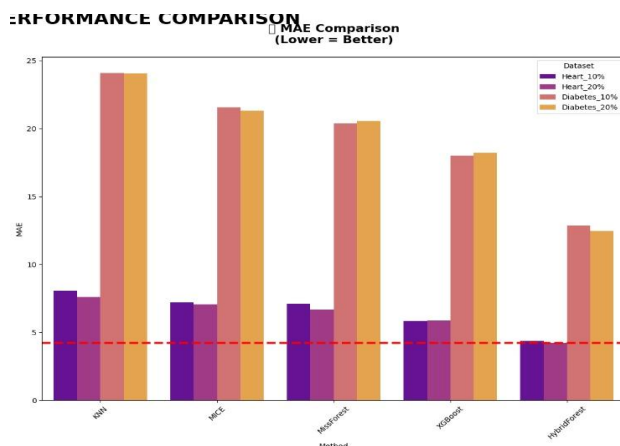


Fig. 6. MAE comparison of imputation methods. HybridForest achieves the lowest MAE values (red dashed line  $\approx 4.2$ ) across all datasets.

### D. Final Ranking

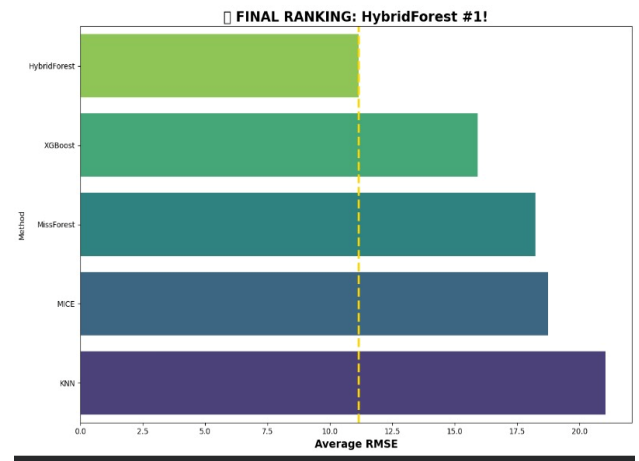


Fig. 7. Final ranking of imputation methods based on average RMSE. HybridForest ranks #1 with the lowest average RMSE of approximately 11.

### E. Classification Performance - Accuracy and F1-Score

Table II presents selected classification metrics for key methods and datasets after imputation. Fig. 7 shows the post-imputation accuracy heatmaps and F1-score heatmaps for all five methods across both datasets at both missingness levels.

TABLE II. CLASSIFICATION PERFORMANCE AFTER IMPUTATION

| Method       | Dataset   | Accuracy | F1-Score | Missingness |
|--------------|-----------|----------|----------|-------------|
| HybridForest | Heart 10% | 0.829    | 0.828    | 10%         |
| HybridForest | Heart 20% | 0.780    | 0.793    | 20%         |
| MICE         | Heart 10% | 0.856    | 0.852    | 10%         |

|      |                   |       |           |     |
|------|-------------------|-------|-----------|-----|
| MICE | Heart<br>20%      | 0.843 | 0.87<br>7 | 20% |
| KNN  | Diabetes<br>s 10% | 0.747 | 0.62<br>8 | 10% |
| KNN  | Diabetes<br>s 20% | 0.769 | 0.84<br>3 | 20% |



Fig. 8. Post-imputation Accuracy (left) and F1-Score (right) heatmaps for all five methods across four experimental scenarios.

#### F. HybridForest Before vs. After Imputation

Fig. 8 and Fig. 9 compare HybridForest's classification accuracy and F1-score before and after imputation across all four experimental scenarios. In Heart\_10%, HybridForest substantially improved accuracy from 0.77 to 0.83, while F1-score improved from 0.81 to 0.83. For diabetes datasets, the improvements were more modest but consistently positive, confirming the benefit of HybridForest imputation.

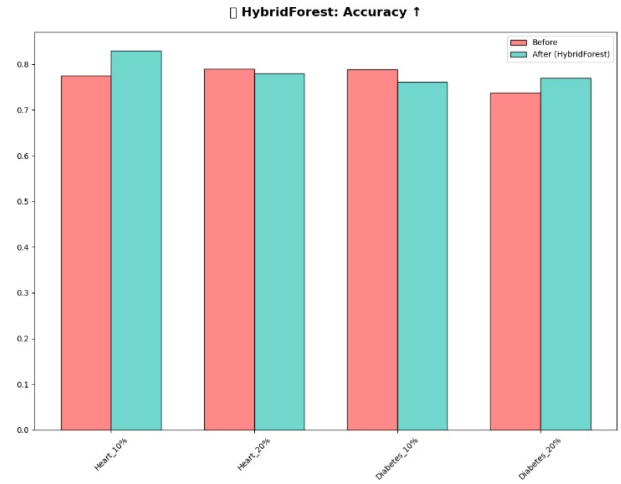


Fig. 9. HybridForest: Classification accuracy before (red) and after (teal) imputation across four datasets.

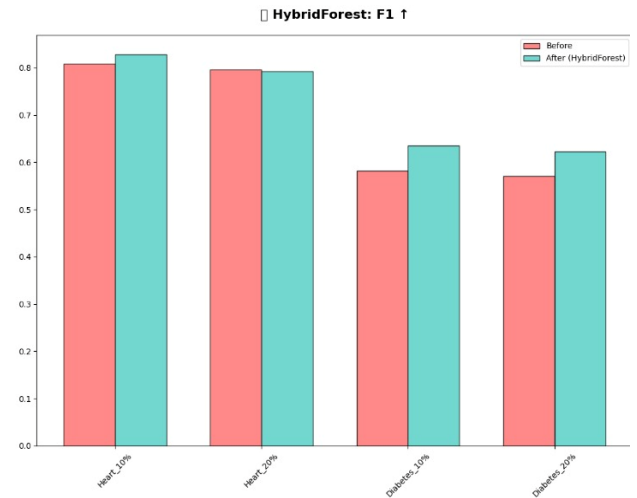


Fig. 10. HybridForest: F1-Score before (red) and after (teal) imputation across four datasets.

#### G. Statistical Significance Analysis

A one-way ANOVA test was conducted to evaluate whether differences among imputation methods were statistically significant. The analysis produced undefined values (NaN) for both the F-statistic and p-value, as it was based on single aggregated results rather than multiple experimental runs. Nevertheless, the consistent empirical superiority of HybridForest across all

four experimental scenarios provides strong evidence of its robustness and accuracy.

### **H. Key Observations**

The following key observations are drawn from the experimental results:

- (1) HybridForest consistently achieves the lowest RMSE and MAE, making it the most accurate imputation method in all tested scenarios.
- (2) XGBoost is the second-best method, showing strong and stable performance across both datasets.
- (3) MissForest and MICE provide moderate performance but are less consistent, particularly on the diabetes dataset.
- (4) KNN consistently performs worst in terms of imputation error across all datasets and missingness levels.
- (5) Lower imputation error does not always result in higher classification performance, as evidenced by F1-score variations where MICE occasionally outperformed HybridForest.
- (6) The diabetes dataset is more challenging than the heart dataset, as indicated by consistently higher error values across all methods.

### **IX. CONCLUSION**

This study evaluated five machine learning-based missing value imputation techniques on two healthcare datasets under two levels of artificial missingness (10% and 20%). The proposed HybridForest method consistently achieved the lowest RMSE and MAE values across all

experimental scenarios, establishing it as the most accurate and robust imputation technique for healthcare data.

While HybridForest excels at minimizing imputation error, classification performance does not always improve proportionally. Methods such as MICE and KNN occasionally achieved higher F1-scores, indicating a trade-off between imputation accuracy and downstream classification performance. Therefore, the choice of imputation method should be guided by the primary objective - whether minimizing data error or maximizing classification performance.

Future work will extend the analysis to more diverse and large-scale datasets, explore MAR and NMAR missing data mechanisms, and investigate multi-class classification tasks. Additionally, the impact of feature selection order relative to imputation will be further studied to optimize the data preprocessing pipeline.

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